Making Taylor expansion compositionnal

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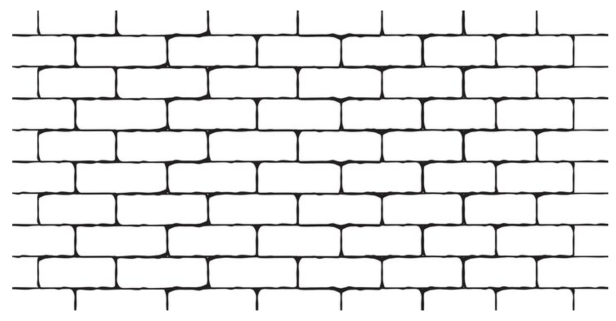
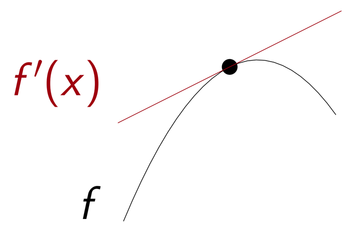
Compositionality

"Composing the meaning of parts into the meaning of the whole."
(Milner)

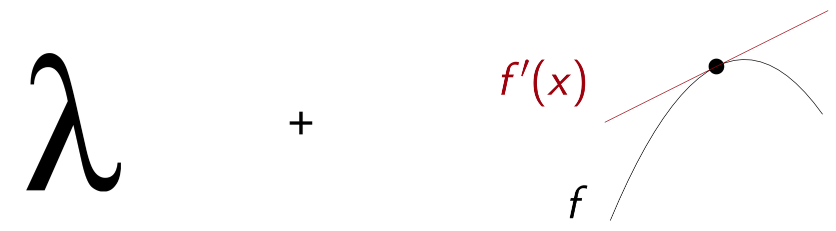


Let us turn **differential calculus** into a compositional theory.



$$f(x+u) = f(x) + f'(x)\cdot u + o(u)$$

Differential calculus and programming languages



- Automated differentiation: derivatives of programs $P : \mathbb{R}^k \to \mathbb{R}$
- Differential λ-calculus: quantitative properties on program execution

Categorical semantics: A framework for compositionality

- correctness of automated differentiation
- models of differential λ-calculus.

But no need for any category theory here.

Making derivatives compositional

Chain rule: bad

$$(g\circ f)'(x) = g'(f(x)) \circ f'(x) \qquad ☹$$

Tangent bundle

$$\frac{f : X \to Y}{\mathrm{T}f : X\times X \to Y \times Y} \qquad \mathrm{T}f(x,u) = (f(x), f'(x)\cdot u)$$

Tangent bundle functoriality: good

$$\mathrm{T}(g\circ f) = \mathrm{T}g \circ \mathrm{T}f \qquad ☺$$

Dual numbers intuition

$(a_0, a_1) \in \mathrm{T}X \iff$ polynomial $a_0 + a_1\varepsilon$ with $\varepsilon^2 = 0$

$\mathrm{T}f$: **pushes forward** polynomials

$$f(a_0 + a_1\varepsilon) = f(a_0) + \varepsilon f'(a_0)\cdot a_1$$

Making Taylor expansion compositional

Faá di Bruno formula: bad (and terrifying)

$$\xcancel{(g\circ f)^{(n)}(x) = \sum_{\{I_1,\dots,I_k\} \text{ partition of } [\![1,n]\!]} g^{(k)}\left(f(x), f^{(I_1)}(x), \dots, f^{(I_k)}(x)\right)} \qquad ☹$$

with $f^{(\{i_1,\dots,i_l\})}(x) : (u_1,\dots,u_n) \mapsto f^{(l)}(x)(u_{i_1},\dots,u_{i_l})$

Taylor functor

$$\mathrm{T}_n X = X \times X^n \qquad \frac{f : X \to Y}{\mathrm{T}_n f : \mathrm{T}_n X \to \mathrm{T}_n Y}$$

Taylor functor: good

$$\mathrm{T}_n(g\circ f) = \mathrm{T}_n g \circ \mathrm{T}_n f \qquad ☺$$

Dual numbers intuitions

$a \in \mathrm{T}_n X \iff$ polynomial $\sum_{k=0}^{n} a_k\, \varepsilon^k$ with $\varepsilon^{n+1} = 0$

$\mathrm{T}_n f$: **pushes forward** polynomials.

- Order n Taylor expansion of $f\left(a_0 + \sum_{k=1}^{n} a_k \varepsilon^k\right)$ at a_0
- Develop by multilinearity
- Simplify by $\varepsilon^{n+1} = 0$.

example: $\mathrm{T}_2 f(x, u_1, u_2) = \left(f(x),\ f'(x)\cdot u_1,\ \frac{1}{2} f^{(2)}(x)(u_1,u_1) + f'(x)\cdot u_2 \right)$

T_n is a **monad** that implements dual numbers

Compositionality and Taylor expansion

I want to combine **Taylor expansion** with **programming languages**

$$f(x+u) = f(x) + \sum_{k=1}^{n} \frac{1}{k!} f^{(k)}(x)(u,\dots,u) + O(u^{n+1})$$

- In AD: higher order derivatives.
- In differential λ-calculus: Taylor expansion of programs.

The strength of category theory

Manipulate power series without ever writing down the power series.

Differential calculus	Category theory
Faá di Bruno	Functoriality of T_n
Dual numbers	Kleisli category of monad T_n
Schwarz theorem	Distributive law $c : \mathrm{T}_n\mathrm{T}_n \Rightarrow \mathrm{T}_n\mathrm{T}_n$
Partial derivatives/Taylor expansion	Monad strength
Multiple dual numbers $\varepsilon_1, \dots, \varepsilon_l$	Canonical monad $\mathrm{T}_{n_1}\cdots\mathrm{T}_{n_l}$

All proof on the category theory world: **no combinatorics**

Central trick: injection

$$\mathrm{L}_n : \mathrm{T}_n X \to \mathrm{T}^n X$$

L_n is a **monad transformer**.

The iterated Tayor expansion

$a \in \mathrm{T}_{n_1}\cdots\mathrm{T}_{n_k} X \iff$ polynomial $\sum_{k_1,\dots,k_l} a_{k_1,\dots,k_l}\, \varepsilon_1^{k_1}\cdots\varepsilon_l^{k_l}$ with $\varepsilon_k^{n_k+1} = 0$

$\mathrm{T}_{n_1}\cdots\mathrm{T}_{n_l} f$: **pushes forward** this polynomial with the Taylor expansion of f.

Infinitary Taylor expansion

Works the same for $n = \infty$, $\mathrm{T}_\infty X = X^{\mathbb{N}}$

Power series intuition

$a \in \mathrm{T}_\infty X \iff$ power series $\sum_{k=0}^{\infty} a_k\, \varepsilon^k$, no equation on ε

$\mathrm{T}_\infty f$: **pushes forward** power series with the Taylor expansion of f

Categorical meaning of analyticity: naturality of $\sigma : (x_0, x_1, \dots) \mapsto \sum_{i=0}^{\infty} x_i$.

$$f \text{ analytic map} \iff \sigma \circ \mathrm{T}_\infty f = f \circ \sigma$$

Taylor expansion is a **bimonad**

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